

Health Professional Copilot

Minimum Viable Product - Product, Safety & Technical Specification

A second layer of attention for every consultation.

Product module: Nutrition Module - AI-assisted pre-consultation support for nutrition professionals.

Platform direction: other profession-specific modules may be added to Health Professional Copilot in the future.

Product Vision

The Nutrition Module helps nutrition professionals arrive at consultations better prepared. It transforms patient-provided intake data, known medical diagnoses, relevant existing laboratory results, medications, and referral context into a structured state; identifies information gaps, nutritional risk factors, and blind spots; proposes nutrition considerations and questions; and retrieves approved supporting evidence.

Medical diagnosis is input context established by an appropriate medical professional, not an output of this module. When the available information suggests that medical assessment may be required, the module raises a referral or escalation flag with a clear recommendation for professional review.

MVP principle: useful suggestions must remain easy to inspect, question, verify, contextualize, reject, and audit.

1. Problem, Users & Outcome

1.1 Problem Statement

Nutrition professionals synthesize dietary patterns, symptoms, goals, medical diagnoses, medications, supplements, lifestyle, relevant existing laboratory results, and clinical guidance under limited time and attention. Traditional intake forms collect information but do not distinguish meaningful absence from missing data, identify what deserves clarification, or highlight when the patient may require medical assessment.

1.2 Intended Users

- Primary MVP user: nutrition professionals conducting initial consultations.
- Patient: completes a concise structured intake before the appointment.
- Referring or treating professional: may provide established diagnoses, referral reason, restrictions, and existing results.
- Clinical/content administrator: approves evidence sources, scope rules, referral/escalation rules, and releases.
- Future professional modules remain supported through module-specific schemas, rules, instructions, and evidence collections.

1.3 Product Objective

Can an evidence-grounded AI agent reduce relevant information gaps and professional blind spots when preparing for a nutrition consultation, while remaining within the nutritionist's scope of practice?

1.4 MVP Success Outcome

A nutrition professional receives a concise, source-linked preparation brief, can distinguish established medical context from nutrition-focused considerations, can inspect supporting evidence, and is alerted when medical referral or escalation may be appropriate.

2. Scope & Safety Boundaries

Capability	MVP status
Summarize patient-provided information	Allowed
Identify missing or ambiguous information	Allowed
Suggest questions for professional review	Allowed
Retrieve approved clinical evidence	Allowed
Surface nutrition considerations	Allowed with evidence status
Identify nutritional risk factors	Allowed
Raise referral or escalation flags	Allowed; deterministic rules first
Use established medical diagnoses as context	Allowed as input only
Form, infer, or present a medical diagnosis	Out of scope
Prescribe or change medication	Out of scope
Order new laboratory tests or contact patients autonomously	Out of scope

2.1 Deterministic Safety Layer

Before agent reasoning, deterministic application logic evaluates configured referral/escalation indicators, contraindications, excluded contexts, critical missing data, and schema validity. The result is one of: continue, continue with a referral/escalation flag, or stop automated analysis and request professional review.

- Rules are versioned, clinically approved, testable, and specific to the Nutrition Module.
- The model may surface additional referral concerns but is not the sole detector.
- A flag states that medical assessment may be required; it never states or implies a new diagnosis.

Fail safe: when safety status, applicability, or evidence support cannot be established, the system abstains and explains why.

3. End-to-End Workflow

Step	Stage	Output
1	Consent & Intake	Patient provides consent and structured pre-consultation data.
2	Patient State	Validated state with explicit unknown and clarification statuses.
3	Scope & Safety Check	Deterministic continue, referral/escalation flag, or stop decision.
4	Nutrition Reasoning Agent	Gaps, nutrition considerations, risks, flags, blind spots, and evidence requests.
5	Evidence Agent	Approved passages, applicability assessment, source metadata, and evidence status.
6	Evidence Gate	Deterministic inclusion, warning, abstention, or rejection.
7	Professional Brief	Concise, source-linked, structured preparation brief.
8	Professional Feedback	Accept, reject, correct, or flag every suggestion.
9	Audit & Evaluation	Versioned trace, outcomes, and quality metrics.

3.1 Two-Agent Architecture

The MVP intentionally retains two meaningful agents. The Nutrition Reasoning Agent determines what deserves attention within nutrition practice and formulates evidence requests. The Evidence Agent searches and assesses approved material. Scope checks, referral rules, evidence gating, rendering, permissions, and audit persistence remain deterministic wherever possible.

Patient Intake -> Patient State -> Scope & Safety Layer -> Nutrition Reasoning Agent -> Evidence Agent -> Evidence Gate -> Professional Brief -> Nutrition Professional

4. Patient Intake & Structured State

4.1 Intake Design

The intake remains intentionally concise and adaptive. It establishes enough context for meaningful preparation without attempting to replace the consultation or a full clinical history.

Area	Representative fields
Identity & context	Age, sex-related clinical context when relevant, height, weight, consultation reason, goals
Symptoms	Relevant symptoms, onset, duration, severity, trajectory, associated features
Known medical context	Established diagnoses, referring professional, referral reason, medications, supplements, allergies, restrictions
Lifestyle	Dietary pattern, activity, sleep, alcohol and smoking when relevant
Existing results	Relevant existing laboratory results with value, unit, date, reference range, and source
Additional context	Recent weight change, family history, pregnancy status when relevant, open notes

4.2 Field-State Contract

Every clinically relevant field is represented as Known, Unknown, Not Applicable, or Requires Clarification. Missing information is never interpreted as a negative finding.

- Each value records provenance: patient-entered, professional-entered, imported, or derived.
- Derived values retain the inputs and transformation version.
- Contradictions remain visible and become Requires Clarification.
- Free text is preserved separately from normalized concepts.

4.3 Consent & Data Minimization

The patient is shown the purpose of processing, intended professional audience, retention policy, and limits of the system. Only information necessary for pre-consultation preparation is collected.

5. Nutrition Reasoning Agent

5.1 Responsibilities

- Read the patient state, including established diagnoses and relevant existing labs.
- Identify nutrition-relevant missing or ambiguous information.
- Generate approximately 3-7 high-value questions or topics.
- Surface nutrition considerations and nutritional risk factors for professional review.
- Raise referral/escalation flags when medical assessment may be required and state the recommended next step.
- Detect blind spots, request evidence for material claims, and abstain outside nutrition scope.

5.2 Structured Output Contract

Object	Required fields
Gap	topic, rationale, priority, source patient fields
Question	topic, proposed wording, rationale, related gap
Nutrition consideration	label, rationale, supporting facts, limiting facts, missing facts, evidence request
Nutritional risk factor	factor, observed context, relevance, priority
Referral/escalation flag	trigger, rationale, urgency, recommendation, supporting evidence/rule
Blind spot	topic, rationale, priority, evidence request when material
Evidence request	clinical question, population/context, claim type, desired source class

5.3 Constraints

- No generation, inference, ranking, or probability of medical diagnoses.
- Known diagnoses are copied from attributed input and clearly labeled as established context.
- No hidden conversion of missing data into negative findings.
- No recommendation to order new laboratory tests; only relevant existing results are summarized.
- Referral flags indicate possible need for medical assessment and do not assert a diagnosis.
- Every output references the patient-state fields that motivated it.

6. Evidence Agent & Knowledge Governance

6.1 Responsibilities

The Evidence Agent searches only approved, versioned collections; evaluates population, nutrition context, and scope applicability; returns source metadata; and assigns an evidence status. It supports nutrition considerations and referral recommendations but does not formulate medical diagnoses or independently add content to the brief.

6.2 Evidence Status

Status	Meaning	Default presentation
Supported	Passage directly supports the material claim in the relevant context.	Eligible for brief
Partially supported	Support exists but applicability or claim scope is incomplete.	Warning or omit
Unsupported	Retrieved material does not support the claim.	Reject
Retrieval failed	No adequate approved passage was found.	Abstain
Outside scope	Population, specialty, or action is outside configured coverage.	Reject and label

6.3 Approved Source Registry

- Owner and clinical approver.
- Title, issuing body, jurisdiction, population, specialty, version, publication date, and review date.
- Evidence hierarchy and allowed claim types.
- Effective and retirement dates, with superseded versions retained for audit.
- Conflict-resolution policy when approved sources disagree.

6.4 Citation Contract

Every evidence item retains the document identifier and version, page or section, exact supporting passage, retrieval score, applicability assessment, ingestion version, and access date.

7. Evidence Gate & Professional Brief

7.1 Evidence Gate

Evidence gating is a product constraint, not a visual feature. Deterministic logic evaluates structured statuses returned by the agents and controls inclusion.

Content type	Minimum gate
Nutrition considerations	Explicit evidence status and applicable nutrition context
Referral/escalation flags	Configured rule or supported evidence, rationale, urgency, and recommended medical follow-up
Material clinical claims	Supported or qualified as partially supported with professional-facing limitation
Questions and information gaps	Patient-state rationale; evidence required when the rationale itself is a material clinical claim
Existing laboratory results	Attributed input only; preserve value, unit, date, range, and source without ordering new tests

7.2 Professional Brief Sections

Section	Purpose
Patient Overview	Consultation reason, goals, symptoms, context, and explicit unknowns
Known Medical Context	Attributed established diagnoses, referral context, medications, restrictions
Information to Clarify	Nutrition-relevant missing, ambiguous, or contradictory information
Suggested Questions	Approximately 3-7 high-value topics for the consultation
Nutrition Considerations	Evidence-statused nutrition-focused considerations for review
Nutritional Risk Factors	Observed factors relevant to nutrition assessment and preparation
Referral / Escalation Flags	Possible need, urgency, rationale, and recommended medical follow-up
Potential Blind Spots	Relevant areas that may otherwise remain unexplored
Supporting Evidence	Approved passages, applicability, limitations, and source metadata
Relevant Existing Labs	Attributed existing results with value, unit, date, range, and source

The brief must distinguish patient facts, established medical context, nutrition-focused AI suggestions, referral flags, retrieved evidence, and deterministic system decisions.

8. Professional Control & Feedback

8.1 Review Actions

The professional remains the final decision-maker and can perform the following action on every generated item:

Action	Meaning
Accept	Useful for consultation preparation; not equivalent to clinical confirmation.
Reject	Not useful or not applicable, with optional reason.
Correct	Clinician supplies corrected wording or context.
Already known	Valid but adds no preparation value.
Evidence problem	Citation is incorrect, weak, outdated, or inapplicable.
Safety concern	Potentially harmful, misleading, or outside scope; triggers review workflow.

8.2 Feedback Use

- Feedback supports evaluation, error analysis, prompt changes, and source governance.
- No automatic learning from production data occurs without approval and de-identification controls.
- High-severity safety feedback is routed for clinical governance review.
- Release comparisons use fixed evaluation sets before production changes.

8.3 User Experience Requirements

- The brief is scannable in under three minutes for typical cases.
- Known Medical Context is visually distinct from Nutrition Considerations.
- Referral / Escalation Flags are prioritized; excessive low-value flags are measured and penalized.
- Relevant Existing Labs never appear as recommendations for new testing.
- The professional can export or record the reviewed brief without representing it as a medical assessment.

9. Privacy, Security & Auditability

9.1 Minimum Controls

- Documented consent, purpose limitation, and data minimization.
- Encryption in transit and at rest; secrets separated from application data.
- Role-based access for patients, professionals, administrators, and clinical reviewers.
- Tenant separation and auditable access to patient records.
- Configurable retention, deletion, and export workflows.
- Model-provider settings that prevent patient data from being used for provider training where applicable.
- Incident logging and a defined escalation path.

9.2 Traceability Record

Persist an auditable execution record without relying on hidden chain-of-thought. Capture versions and observable events required to reproduce and investigate the result:

- Consent and structured input version.
- Safety rule-set version and outcomes.
- Agent instructions, model and configuration version, structured requests and structured results.
- Tool calls, retrieved passages, evidence statuses, retries, timeouts, and gate decisions.
- Brief version, professional feedback, timestamps, latency, and cost.

Auditability means reproducing the inputs, observable decisions, evidence, and released output - not storing private model reasoning.

10. Validation Dataset & Method

10.1 Evaluation Dataset

Use fixed synthetic patient cases to protect privacy and ensure reproducibility. Clinical ground truth is created from real approved sources and predefined before either the baseline or agentic system is evaluated.

- Include straightforward, missing-information, multiple-nutrition-consideration, distractor, insufficient-evidence, blind-spot, referral-needed, urgent-escalation, known-diagnosis, no-diagnosis, contradictory-data, retrieval-failure, and negative-control cases.
- Maintain separate development and locked test sets.
- Represent expected concepts semantically, not through exact wording.
- Trace expected nutrition considerations and referral flags to approved sources or deterministic rules.
- Use qualified nutrition reviewers, with medical review of referral/escalation cases, and adjudicate disagreements.

10.2 Comparison Arms

System	Purpose
Intake only	Measures the benefit of any generated preparation support.
Single-prompt baseline	Patient Intake -> LLM -> Brief, without tools, retrieval, or agent loop.
Two-agent MVP	Nutrition Reasoning Agent + Evidence Agent + deterministic scope, referral, and evidence gates.
Ablations	Remove one component at a time to measure its contribution.

11. Metrics & Initial Acceptance Targets

Targets below are provisional engineering release gates. They must be reviewed with clinical evaluators after a pilot dataset is established; failing a safety gate blocks release.

Metric	Initial MVP target
Safety violation rate	0 critical violations on locked test set
Referral flag recall	>= 95%; 100% on configured urgent escalation cases
Referral flag precision	>= 85% on locked test set
Citation correctness	>= 95% of displayed material claims
Evidence applicability	>= 90% judged applicable by reviewers
Nutrition consideration recall	>= baseline + 15 percentage points
Relevant information gap recall	>= baseline + 15 percentage points
Irrelevant question rate	<= 15%
Structured-output validity	>= 99% after one retry
Typical brief size	3-7 questions; <= 5 nutrition considerations unless referral state
End-to-end latency	p95 <= 20 seconds in evaluation environment
Professional review time	Median <= 3 minutes in formative usability test

11.1 Measurement Rules

- Report precision and recall together; do not optimize recall by over-generation.
- Measure nutrition consideration recall and referral flag recall separately.
- Measure citation correctness separately from retrieval similarity.
- Count abstention as correct when evidence is insufficient or a request falls outside nutrition scope.
- Stratify results by case type and referral urgency; averages must not hide critical failures.
- Record confidence intervals and reviewer agreement when the sample permits.

12. Technical Architecture

Layer	MVP responsibility
Frontend	Patient intake; professional dashboard; evidence and feedback interactions
API/backend	Authentication, workflows, structured models, deterministic validation and gates
Nutrition Reasoning Agent	Gaps, nutrition considerations, risk factors, referral flags, blind spots, evidence requests
Evidence Agent	Approved-source retrieval, applicability assessment, source metadata
Knowledge pipeline	Extraction, chunking, indexing, versioning, approval and retirement
Storage	Relational patient/workflow data plus evidence index and immutable audit events
Observability	Latency, cost, failures, retrieval, gate decisions, feedback, evaluation
Security	Identity, access control, encryption, tenant isolation, retention and deletion

12.1 Reliability Requirements

- Typed schemas at every agent and tool boundary.
- Established diagnoses and existing lab results retain input provenance and are never model-inferred.
- Idempotent workflow steps and explicit timeout/retry limits.
- No silent fallback from approved retrieval to general model knowledge.
- If the Evidence Agent fails, the brief records the failure and suppresses gated content.
- Version every prompt, model, rule set, source collection, schema, and release.
- Do not log unnecessary patient free text in operational telemetry.

12.2 Primary Failure Modes

- Missed referral: available information indicates need for medical assessment but no flag appears.
- Over-referral: low-specificity flags create alert fatigue and unnecessary escalation.
- Scope leakage: model language implies or generates a medical diagnosis.
- Retrieval miss or citation mismatch: applicable support is absent or does not justify the displayed item.
- Applicability error: guidance is valid but not for this nutrition context.
- Automation bias: nutrition suggestions appear more authoritative than intended.

13. Delivery Plan & Definition of Done

Phase	Deliverables
Phase 1 - Core	Consent/intake, patient state, professional brief, single-prompt baseline, audit skeleton
Phase 2 - Two agents	Nutrition Reasoning Agent, Evidence Agent, approved source registry, citations
Phase 3 - Reliability	Scope/safety layer, referral rules, Evidence Gate, feedback, locked evaluation
Phase 4 - Pilot readiness	Security review, clinical content approval, usability test, operating runbook
Post-MVP	Longitudinal context, integrations, additional professional modules, prospective validation

13.1 MVP Definition of Done

- End-to-end workflow runs for the nutrition consultation intake without manual data repair.
- Both agents return valid structured outputs and failures degrade safely.
- Medical diagnoses appear only when supplied as attributed input; generated outputs contain no diagnostic inference.
- Relevant existing labs preserve values and provenance; the module does not recommend new tests.
- Configured referral/escalation rules pass the locked suite and referral recall target.
- Nutrition considerations and material claims satisfy their evidence gates.
- Every brief is reproducible from versioned inputs, evidence, rules, and configuration.
- Feedback, privacy controls, and acceptance targets are tested and reported.
- Nutrition and medical governance owners approve sources, scope rules, referral rules, limitations, and pilot release.

The MVP is complete when it demonstrates safer, traceable preparation value under measurable constraints - not when it merely produces plausible clinical text.

14. Product Perspective

The Nutrition Module supports nutrition practice; it does not attempt to replace medical diagnosis.

Medical diagnoses enter as attributed patient or professional context. The module adds value by improving preparation, surfacing nutrition-relevant gaps and risks, organizing existing information, and identifying when medical assessment may be appropriate.

Referral and escalation flags are prompts for professional judgment. They communicate a possible need for medical evaluation and a recommended next step without naming or implying a new diagnosis.

Explicit MVP Decisions

- This application is the Nutrition Module of Health Professional Copilot.
- The MVP intentionally retains two agents: Nutrition Reasoning and Evidence.
- Medical diagnosis is input context; diagnostic generation is outside scope.
- The module can raise referral/escalation flags and recommend medical assessment when appropriate.
- Other professions and modules may be added to the platform in the future.

A second layer of attention for every consultation.